Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - II

Course Code: BCA104-2 **Max Marks: 100**

Course Name: DISCRETE MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Demonstrate a working knowledge of set notation and elementary set theory and recognize the connection between set operations and logic.

CO2: Apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3: Understand different types of matrices and operations among matrices.

CO4: Determine when a relation is reflexive, symmetric, antisymmetric, transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5: Effective express the elementary concepts and results of Number theory.

Section A **3*8=24**

1)	Using set notation, prove that if A B and B C, then A C. (OR) Define the Cartesian product of two sets and explain its significance in discrete mathematics.	**3**	(CO1, L4)
2)	Describe the properties of a function f: A → B that is both injective and surjective. (OR) Explain how to determine if a given relation on a set is reflexive, symmetric, or transitive.	**3**	(CO2, L4)
3)	Prove that the composition of two injective functions is also an injective function. (OR) Define a matrix and explain its role in solving systems of linear equations.	**3**	(CO2, L4)
4)	Find the inverse function f^(-1) for a given bijective function f: A → B. (OR) Define and illustrate an equivalence relation, providing an example of its application.	**3**	(CO2, L4)
5)	Illustrate the concept of a partial ordering on a set, and give an example. (OR) Explain the difference between a matrix and a list in terms of their use in discrete mathematics.	**3**	(CO4, L2)
6)	Determine if the relation R on the set of integers Z defined by (x, y) R if and only if x - y is even is an equivalence relation. (OR) Prove that every integer has a unique prime factorization.	**3**	(CO4, L4)
7)	Define the set of natural numbers N and explain its significance in number theory. (OR) Provide an example of a surjective function that is not injective.	**3**	(CO5, L2)
8)	Explain the principle of mathematical induction and demonstrate its use in proving a statement. (OR) Describe the properties of symmetric matrices.	**3**	(CO4, L4)

AI-assisted draft - teacher review and approval required before use

Section B

8*7=56

1) a)	Prove that for any sets A, B, and C, if A ∩ B = A ∩ C, then B = C. (OR) Given the function f(x) = x^2 - 4x + 3, find its inverse function.	**8**	(CO1, L4)
2) a)	Define a matrix and provide an example of a matrix that is both symmetric and skew-symmetric. (OR) Discuss the properties of composition in relation to functions, providing examples.	**8**	(CO3, L3)
3) a)	Given a set A = {1, 2, 3, 4}, define an equivalence relation R on A and describe the partition of A induced by R. (OR) Prove that for any integer n, if n^2 is even, then n must be even.	**8**	(CO4, L4)
4) a)	Construct a truth table for the logical statement (p q) ¬r. (OR) Provide an example of a relation that is both reflexive and transitive but not symmetric.	**8**	(CO1, L3)
5) a)	Define the concept of a matrix transpose and show that (AB)^T = B^T A^T. (OR) Explain how to determine if a given relation is an equivalence relation or a partial ordering.	**8**	(CO3, L4)
6) a)	Describe the process of Gaussian elimination and its application in solving a system of linear equations. (OR) Provide an example of a function that is surjective but not injective.	**8**	(CO3, L3)
7) a)	Explain the concept of a prime number and its role in modern cryptography. (OR) Describe the properties of matrix addition and multiplication.	**8**	(CO5, L3)

Section C

10*2=20

1) a)	Given two sets A = {1, 2, 3} and B = {4, 5}, construct the Cartesian product of A and B. (OR) Provide a detailed explanation of the principle of mathematical induction.	**10**	(CO1, L4)
2) a)	Define and illustrate an equivalence relation on the set of integers Z. Then, describe the partition of Z induced by this relation. (OR) Explain how to determine if a given function is bijective.	**10**	(CO4, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

AI-assisted draft - teacher review and approval required before use